

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE) *IT*

May 2012

CE204 Digital Electronics

Date: 03.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

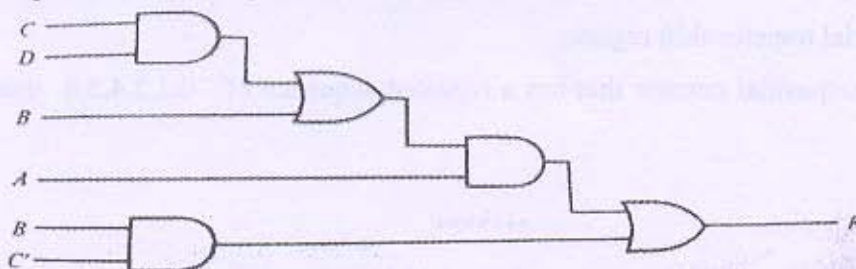
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Prove basic theorems of Boolean Algebra. [04]
(b) Express the Boolean function $F = xy + x'z$ in a product of maxterm form. [03]
- Q - 2 (a) Simplify the Boolean function $F = A'B(D'+C'D) + B(A + A'CD)$ with minimum number of minterms. [04]
(b) Simplify following Boolean function $F(x,y,z) = \sum(0,1,3,5,6)$ using K-map. [04]
(c) Explain full adder in detail and Draw full adder circuit using 2 half adder and OR gate. [06]

OR

- Q - 2 (a) Simplify the Boolean function $F(w,x,y,z) = \sum(1,3,7,11,15)$ and don't care conditions $d(w,x,y,z) = \sum(0,2,5)$ and draw the logic diagram. [06]
(b) Implement following circuit using only NAND gates. [04]



- (c) Explain 3 to 8 decoder in detail. [04]
- Q - 3 (a) Design a combinational circuit for BCD to Excess -3 code conversion. [07]
(b) Explain 4-bit magnitude comparator and draw the diagram of 4-bit magnitude comparator. [07]

OR

- Q - 3 (a) Implement the following function using 8x1 multiplexer : $F(A,B,C,D) = \sum(0,1,3,4,8,9,15)$. [05]
(b) Design a combinational circuit whose input is 3-bit binary number and whose output is (4-bit) 2's complement of the input number. [09]

SECTION - II

Q - 4 (a) Subtract 14 from 25 using 8-bit 1's complement arithmetic. [03]

(b) Convert $(110101.101010)_2$ to Octal. [02]

(c) Convert $(5497)_{10}$ to Binary. [02]

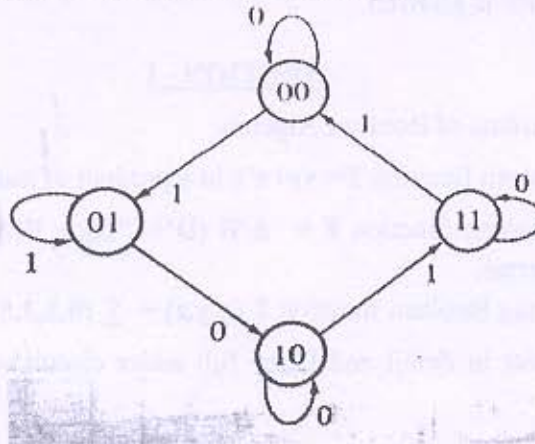
Q - 5 (a) Explain BCD Ripple counter. [07]

(b) Write a Technical note on Master-Slave flip flop. [07]

OR

Q - 5 (a) Describe Johnson Counter. [07]

(b) Design clocked sequential circuit with state diagram shown below using J-K flip flop. [07]



Q - 6 (a) Design a serial adder using sequential logic procedure. [07]

(b) Write a note on memory unit. [07]

OR

Q - 6 (a) Design serial transfer shift register. [07]

(b) Design a sequential counter that has a repeated sequence of 0,1,2,4,5,6 using J-K flip flop. [07]
